Comparative Analysis of Machine Learning Models for Predicting Huntington's Disease Stage

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***Abstract*—This study presents a comparative analysis of various machine learning models applied to a structured dataset for predicting the stage of Huntington's disease. The dataset comprises clinical, genetic, and demographic attributes, including age, sex, family history, HTT CAG repeat length, motor symptoms, cognitive decline, chorea score, brain volume loss, functional capacity, gene mutation type, gene expression level, protein aggregation level, and genomic annotations. A thorough preprocessing phase was undertaken to remove irrelevant features and transform the remaining variables through appropriate encoding and scaling procedures. Feature selection was performed using a correlation based method as well as statistical techniques involving Chi Square and ANOVA to identify the most relevant attributes. The performance of classifiers, including Logistic Regression, Naive Bayes, Support Vector Machine, K Nearest Neighbors, K Means, Decision Tree, Random Forest, and Gradient Boosting, was evaluated across three experimental scenarios: before feature selection, after feature selection using correlation methods, and after feature selection using Chi Square and ANOVA methods. Although overall prediction accuracy remained modest, logistic regression emerged as the best performing model when combined with statistical feature selection. The study highlights the challenges of predicting disease stage due to subtle discriminative patterns in the data and underscores the potential for advanced feature engineering and model tuning in future research.**

***Index Terms*—machine learning, classification, huntington's disease, feature selection, logistic regression, predictive modelling, medical research**

I. Introduction

Huntington's disease is a progressive neurodegenerative disorder that originates from a genetic mutation in the HTT gene. It is typically marked by motor dysfunction, cognitive decline, and psychiatric symptoms. The condition follows an autosomal dominant pattern of inheritance, and its onset and progression are closely associated with the expansion of CAG trinucleotide repeats within the HTT gene. Accurate and timely prediction of the disease stage is essential for effective clinical management, therapeutic planning, and improving the quality of life for individuals affected by the disorder.

The dataset used in this study comprises structured clinical, genetic, and demographic information from patients diagnosed with Huntington's disease. It includes variables such as age, sex, family history, CAG repeat length, motor and cognitive symptoms, and imaging-based indicators like brain volume loss. Additionally, it features molecular-level data such as gene expression, protein aggregation, mutation types, and biological markers relevant to disease mechanisms. These diverse attributes enable a comprehensive understanding of the disease and support the application of machine learning methods for predictive modeling of disease stages, categorized as Early, Pre-symptomatic, Middle and Late.

The primary objective of this study is to conduct a comparative analysis of several machine learning algorithms to evaluate their effectiveness in predicting the stage of Huntington's disease. The models implemented in this study include Logistic Regression, Naive Bayes, Support Vector Machine, K Nearest Neighbors, K Means clustering, Decision Tree, Random Forest, and Gradient Boosting. Model performance is examined across three phases: using the complete feature set, after applying correlation-based feature selection, and after implementing feature selection through a combination of Chi squared and ANOVA F test methods. Through this comparative analysis, the study aims to determine the most effective and interpretable model for the prediction of disease progression in Huntington's disease.

II. Methodology

A. Data Preprocessing

To prepare the dataset for machine learning analysis, several preprocessing steps were undertaken to improve data quality and ensure consistency across features. First, non-informative and irrelevant columns were removed. These included Patient ID, Random Protein Sequence, and Random Gene Sequence, which did not contribute meaningful information for prediction and were excluded to reduce redundancy and dimensionality.

A key focus of the preprocessing involved the Cognitive Decline feature, which contains four categorical levels of impairment: None, Mild, Moderate, and Severe. During initial inspection, it was observed that some instances labeled as None were incorrectly interpreted as missing values due to being stored as empty strings, numerical zeroes, or as the string 'nan'. To address this, all entries were converted to lowercase text and standardized. Subsequently, all placeholder entries representing None were explicitly mapped, and the entire feature was numerically encoded using an ordinal scale where 0 represented None, 1 represented Mild, 2 represented Moderate, and 3 represented Severe. This ensured accurate representation of cognitive decline severity for model training.

In addition, Motor Symptoms, a clinical measure of motor impairment, was also encoded using an ordinal mapping that assigned values from 0 to 3 corresponding to increasing levels of severity. Nominal categorical features including Sex, Family History, Gene Mutation Type, Gene or Factor, Chromosome Location, Function, Effect, and Category were converted into numeric form using label encoding. This transformation allowed the models to interpret and process these categorical inputs effectively.

The target variable, Disease Stage, was also label encoded to convert the categorical labels into numeric values. Specifically, Early was encoded as 0, Late as 1, Middle as 2, and Pre symptomatic as 3. This encoding preserved the distinct progression categories of the disease while making them suitable for classification algorithms.

Finally, continuous numerical features such as Age, HTT CAG Repeat Length, Chorea Score, Brain Volume Loss, Functional Capacity, HTT Gene Expression Level, and Protein Aggregation Level were standardized to have zero mean and unit variance. This normalization ensured that each numerical feature contributed equally during model training, especially for algorithms sensitive to the scale of input variables.

B. Model Implementation

In order to evaluate the predictive performance of various machine learning models for Huntington disease stage classification, all models were implemented in three separate phases. The first implementation was performed on the full feature set before any feature selection. The second implementation followed the application of a variance threshold and correlation-based feature elimination. The third implementation was conducted after applying statistical selection techniques using the chi square and ANOVA F-test, both applied independently to the original unfiltered dataset. This allowed for a comparative assessment of how different feature selection strategies impacted each model's classification accuracy and overall effectiveness.

Eight supervised and unsupervised models were evaluated in total. These included logistic regression, naive Bayes, support vector machine, k nearest neighbors, k means clustering, decision tree, random forest, and gradient boosting. For classification models, the primary evaluation metrics included accuracy score, precision, recall, and F1 score, which were obtained using a classification report. Additionally, confusion matrices were used to visually assess misclassification patterns. For the k means clustering model, cluster labels were aligned to ground truth labels using a majority voting method based on the training set.

Logistic regression was configured with a maximum of one thousand iterations and trained using scaled features. The support vector machine model employed a radial basis function kernel. The k nearest neighbors model was initialized with five neighbors, and feature scaling was applied prior to training. The k means model used the number of unique class labels to determine the number of clusters, and scaling was applied before fitting. For naive Bayes, a Gaussian distribution assumption was used. The decision tree classifier was trained using the default settings with a fixed random state to ensure reproducibility. The random forest classifier was trained with one hundred estimators and a fixed random state. Finally, the gradient boosting classifier was also configured with a fixed random state and trained using its default parameter values.

This multi-phase approach enabled an in-depth comparison of model behavior both with and without feature selection, providing a basis for analyzing how dimensionality reduction and statistical filtering influenced model performance in the classification of Huntington disease progression stages.

C. Feature Selection

To improve model performance and reduce dimensionality, feature selection techniques were employed in two phases. The first phase involved the application of a variance threshold method followed by a correlation analysis. The second phase utilized statistical techniques, specifically the chi square test for categorical variables and the ANOVA F-test for numerical variables. These steps helped identify and eliminate features that contributed little to the prediction of the disease stage, thereby simplifying the model and potentially improving its generalization capability.

1) Variance Threshold and Correlation Analysis: The variance threshold method was used to identify features with low variability across the dataset. The underlying concept of this method is that features with little to no variance provide limited discriminatory power for classification and can therefore be removed. A threshold value of 0.05 was used, meaning any feature with variance below this value would be considered for elimination. However, all features surpassed the threshold, and hence none were removed at this stage. The retained features included: Age, Sex, Family History, HTT CAG Repeat Length, Motor Symptoms, Cognitive Decline, Chorea Score, Brain Volume Loss, Functional Capacity, Gene Mutation Type, HTT Gene Expression Level, Protein Aggregation Level, Disease Stage, Gene or Factor, Chromosome Location, Function, Effect, and Category.

Following this, a correlation matrix was generated to examine the linear relationships among features. The goal was to identify pairs of features with very strong correlations, as these could introduce redundancy into the model. A correlation coefficient threshold of 0.85 was selected. Features showing absolute correlations above this threshold were considered redundant. Based on this analysis, the features Effect and Category were identified as highly correlated and were subsequently removed. The resulting filtered dataset was then used for model implementation to observe changes in performance.

2) Chi Square Test and ANOVA F-test: In the second part of feature selection, statistical methods were applied to the original dataset with all preprocessed features. The chi square test was used to evaluate the relationship between categorical features and the target variable, Disease Stage. This test assesses whether distributions of categorical variables differ from one another significantly. A MinMax scaling was first applied to transform the data to a non negative range, as

required by the chi square test. The results indicated that the features Sex, Effect, Motor Symptoms, Gene or Factor, and Chromosome Location exhibited weak associations with the target variable. These features were removed from the dataset.

TABLE I
CHI2 SCORES FOR DIFFERENT FEATURES

Feature	Chi2 Score
Family_History	4.739131
Gene_Mutation_Type	1.838207
Cognitive_Decline	1.529657
Category	1.087005
Function	1.087005
Sex	1.061497
Effect	0.844298
Motor_Symptoms	0.696346
Gene/Factor	0.539242

For numerical features, the ANOVA F-test was applied to assess the extent to which each feature separated the groups defined by the Disease Stage. This test compares the variance between groups with the variance within groups. The results revealed that Chorea Score and HTT Gene Expression Level had relatively low F scores and thus weak association with the target. These two features were also removed from the dataset.

TABLE II
ANOVA F-TEST SCORES

Feature	F Score
Protein_Aggregation_Level	2.789878
HTT_CAG_Repeat_Length	2.636074
Brain_Volume_Loss	2.097959
Functional_Capacity	1.117649
Age	1.096190
Chorea_Score	0.700453
HTT_Gene_Expression_Level	0.403625

Following the removal of all identified weak features, a final dataset was constructed and used for a third round of model implementation. This comprehensive feature selection process ensured that the retained features contributed meaningful information to the classification task, thereby supporting the goal of building efficient and interpretable models for predicting the progression stage of Huntington's disease.

III. RESULTS

A. Before Feature Selection

This stage involved evaluating machine learning models on the full set of features after data preprocessing but before applying any feature selection techniques. The classification was performed using eight different algorithms, and the results are presented below in terms of precision, recall, F1-score, and overall accuracy. The performance was evaluated for each of the four disease stages, with the stages numerically represented as follows: Middle: 2, Pre-symptomatic: 3, Early: 0, Late: 1

1) Naive Bayes: The Naive Bayes model produced relatively poor performance across all stages, showing a slight bias toward predicting the middle stage (2). The overall accuracy was approximately 25.4%, and class-wise metrics remained low, indicating limited discriminative power in the unfiltered feature set.

- **Accuracy**: 0.254
- **Macro Avg F1-score**: 0.25

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Naive Bayes Classification Report:
              precision    recall  f1-score   support

           0       0.27      0.17      0.21      2457
           1       0.25      0.20      0.22      2394
           2       0.26      0.37      0.30      2456
           3       0.24      0.29      0.26      2401

    accuracy                           0.25      9708
   macro avg       0.26      0.25      0.25      9708
weighted avg       0.26      0.25      0.25      9708

Accuracy: 0.2542233209723939
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Fig. 1. Classification report for Naive Bayes

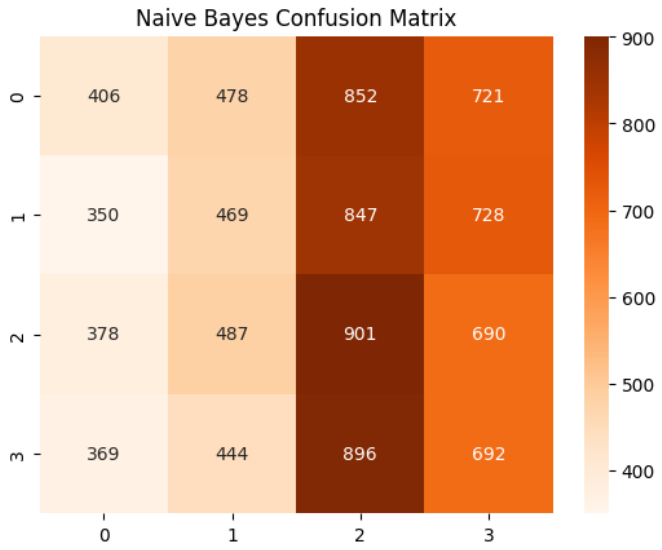


Fig. 2. Confusion Matrix for Naive Bayes

2) Decision Tree: The Decision Tree classifier achieved a similar performance to Naive Bayes, with consistent but modest precision and recall across all classes. The model displayed no significant overfitting but also lacked strong predictive capability.

- **Accuracy**: 0.250
- **Macro Avg F1-score**: 0.25

Decision Tree Classification Report:

	precision	recall	f1-score	support
0	0.26	0.26	0.26	2457
1	0.23	0.23	0.23	2394
2	0.26	0.26	0.26	2456
3	0.25	0.25	0.25	2401
accuracy			0.25	9708
macro avg	0.25	0.25	0.25	9708
weighted avg	0.25	0.25	0.25	9708

Accuracy: 0.2504120313143799

Fig. 3. Classification report for Decision Tree

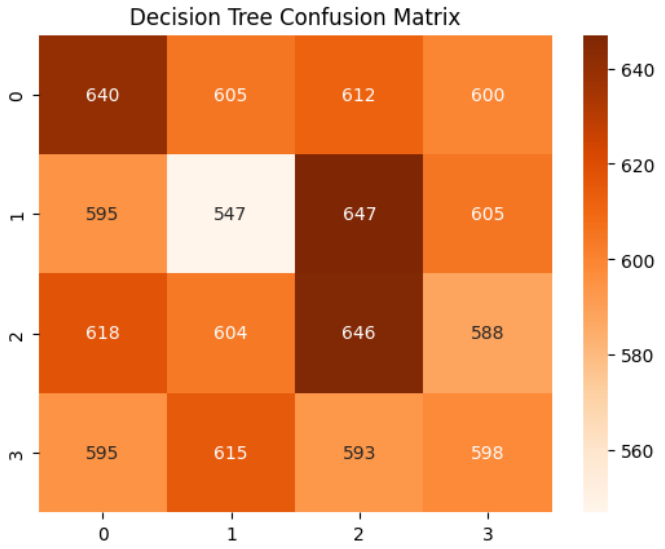


Fig. 4. Confusion Matrix for Decision Tree

3) Random Forest: The Random Forest model slightly improved upon the Decision Tree, achieving an accuracy of 25.7%. The metrics remained consistent across classes, but the overall performance still indicated limited class separation.

- **Accuracy**: 0.258
- **Macro Avg F1-score**: 0.26

Random Forest Classification Report:

	precision	recall	f1-score	support
0	0.26	0.26	0.26	2457
1	0.26	0.25	0.25	2394
2	0.25	0.26	0.25	2456
3	0.26	0.26	0.26	2401
accuracy			0.26	9708
macro avg	0.26	0.26	0.26	9708
weighted avg	0.26	0.26	0.26	9708

Accuracy: 0.257622579316028

Fig. 5. Classification report for Random Forest

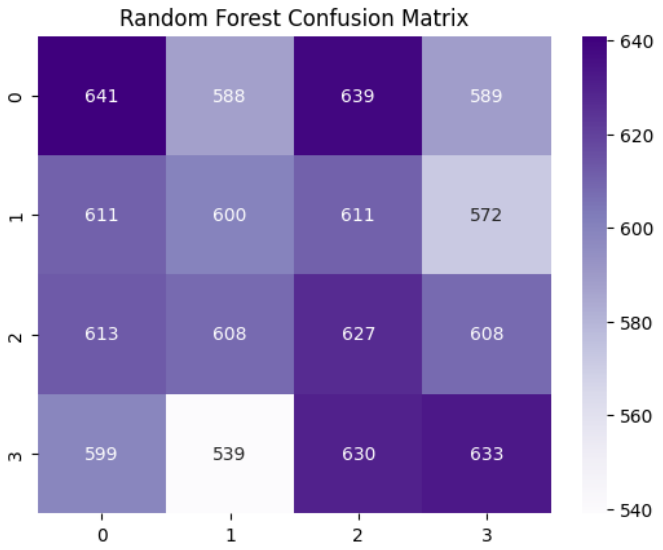


Fig. 6. Confusion Matrix for Random Forest

4) Gradient Boosting: Gradient Boosting struggled with class 0 and class 1 predictions but performed marginally better for class 2 and class 3. The model attained an accuracy of 24.8%, the lowest among ensemble methods at this stage.

- **Accuracy**: 0.248
- **Macro Avg F1-score**: 0.25

Gradient Boosting Classification Report:

	precision	recall	f1-score	support
0	0.25	0.18	0.21	2457
1	0.25	0.22	0.23	2394
2	0.26	0.30	0.28	2456
3	0.24	0.30	0.27	2401
accuracy			0.25	9708
macro avg	0.25	0.25	0.25	9708
weighted avg	0.25	0.25	0.25	9708

Accuracy: 0.2481458590852905

Fig. 7. Classification report for Gradient Boosting

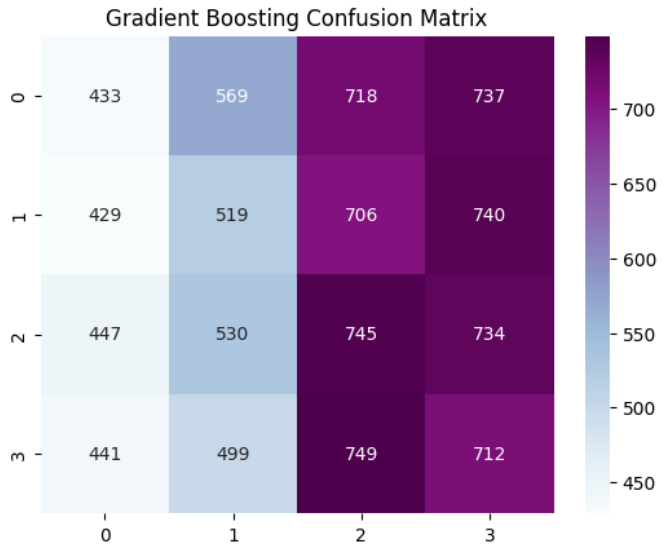


Fig. 8. Confusion Matrix for Gradient Boosting

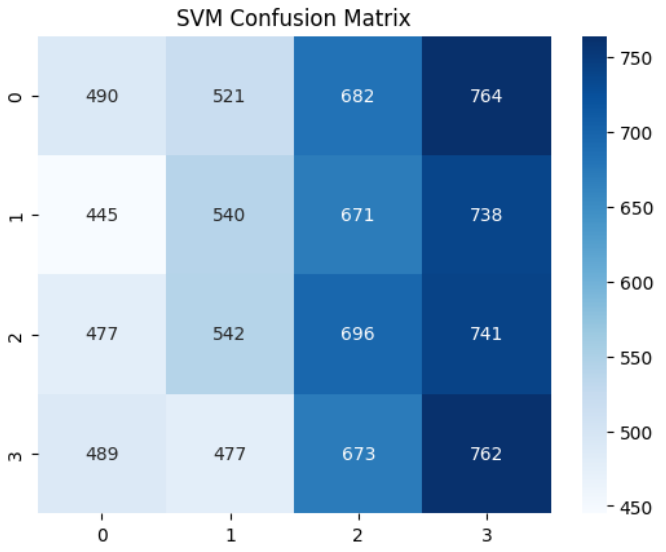


Fig. 10. Confusion Matrix for Support Vector Machine (SVM)

5) Support Vector Machine (SVM): The SVM classifier, using a radial basis function kernel, achieved an accuracy of 25.6%. It showed slightly better recall for class 3 (pre-symptomatic), suggesting its potential to distinguish more advanced disease stages.

- **Accuracy**: 0.256
- **Macro Avg F1-score**: 0.25

SVM Classification Report:

	precision	recall	f1-score	support
0	0.26	0.20	0.22	2457
1	0.26	0.23	0.24	2394
2	0.26	0.28	0.27	2456
3	0.25	0.32	0.28	2401
accuracy			0.26	9708
macro avg	0.26	0.26	0.25	9708
weighted avg	0.26	0.26	0.25	9708

Accuracy: 0.25628347754429337

Fig. 9. Classification report for Support Vector Machine (SVM)

6) Logistic Regression: Logistic Regression offered balanced performance, achieving the joint-highest accuracy of 25.8% at this stage. It was particularly better at recalling pre-symptomatic (3) and middle stage (2) cases, although early and late stages were predicted with less accuracy.

- **Accuracy**: 0.258
- **Macro Avg F1-score**: 0.25

Logistic Regression Classification Report:

	precision	recall	f1-score	support
0	0.28	0.20	0.23	2457
1	0.25	0.17	0.20	2394
2	0.26	0.30	0.28	2456
3	0.25	0.36	0.30	2401
accuracy			0.26	9708
macro avg	0.26	0.26	0.25	9708
weighted avg	0.26	0.26	0.25	9708

Accuracy: 0.257622579316028

Fig. 11. Classification report for Logistic Regression

7) KMeans Clustering: As an unsupervised method, KMeans performed poorly on classes 0 and 1, with the model predominantly predicting class 2. Despite post-hoc label alignment, the overall accuracy remained at 25.1%, and the macro average metrics were notably lower than other models.

- **Accuracy**: 0.251
- **Macro Avg F1-score**: 0.16

8) K-Nearest Neighbors (KNN): KNN achieved one of the better results among the models, particularly for class 0 (early stage), with a relatively high recall of 0.34. However, it underperformed on the pre-symptomatic stage. The overall accuracy was 25.5%.

- **Accuracy**: 0.255
- **Macro Avg F1-score**: 0.25

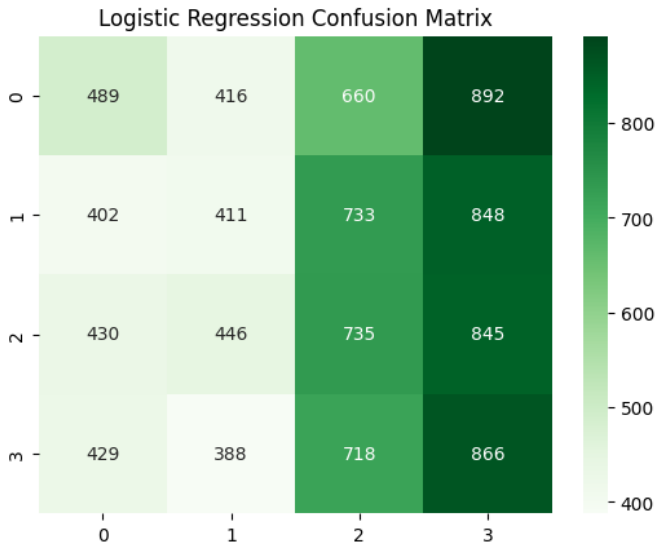


Fig. 12. Confusion Matrix for Logistic Regression

KMeans Classification Report:

	precision	recall	f1-score	support
0	0.00	0.00	0.00	2457
1	0.00	0.00	0.00	2394
2	0.25	0.75	0.38	2456
3	0.24	0.25	0.25	2401
accuracy			0.25	9708
macro avg	0.12	0.25	0.16	9708
weighted avg	0.12	0.25	0.16	9708

Accuracy: 0.25103007828594975

Fig. 13. Classification report for KMeans Clustering

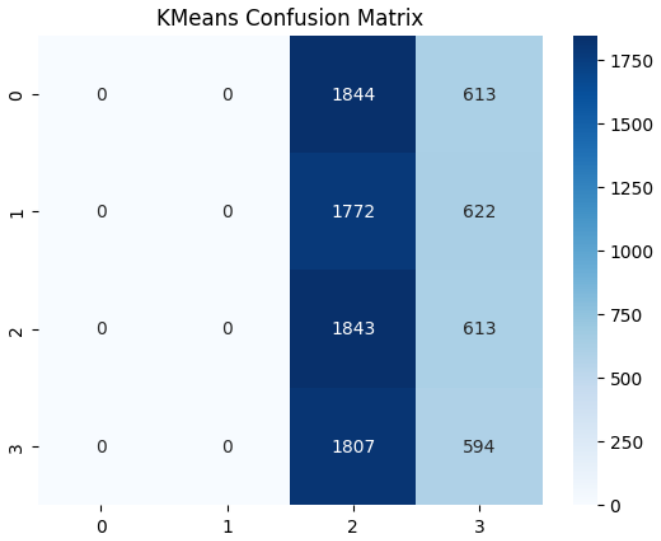


Fig. 14. Confusion Matrix for KMeans Clustering

KNN Classification Report:

	precision	recall	f1-score	support
0	0.26	0.34	0.30	2457
1	0.25	0.28	0.26	2394
2	0.26	0.23	0.24	2456
3	0.25	0.17	0.20	2401
accuracy			0.26	9708
macro avg	0.25	0.25	0.25	9708
weighted avg	0.25	0.26	0.25	9708

Accuracy: 0.25515039142974866

Fig. 15. Classification report for K-Nearest Neighbors (KNN)

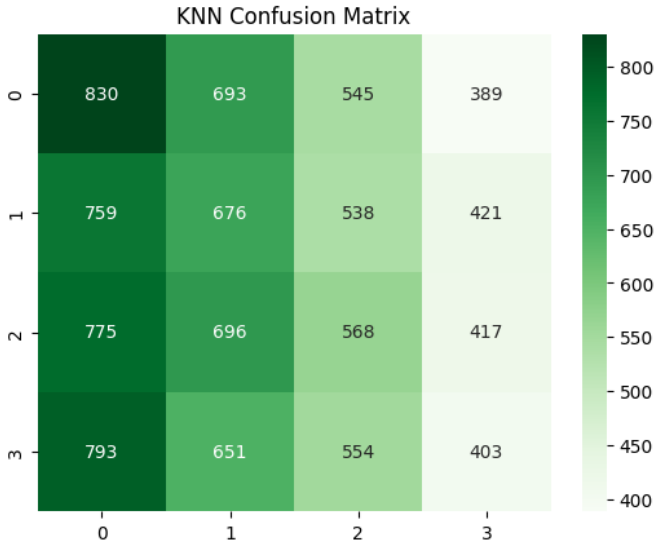


Fig. 16. Confusion Matrix for K-Nearest Neighbors (KNN)

B. After Correlation-based Feature Selection

The correlation method aimed to remove highly correlated features, reducing redundancy and potential multicollinearity in the dataset. However, the impact of this method was mixed across different models.

1) Improved Models::

- Logistic Regression showed a slight increase in performance (from 0.2576 to 0.2579), suggesting that reducing correlated features helped simplify the model without harming its predictive power.
- Naive Bayes, KMeans, and KNN also experienced minor improvements, indicating that these models might have benefited from a leaner feature space where noisy or redundant features were removed.

2) Declined Models::

- Random Forest, Gradient Boosting, and SVM saw a drop in accuracy. These models often handle high-dimensional data well, and removing features based purely on correlation may have eliminated variables carrying valuable predictive information for them.

- For example, Random Forest dropped from 0.2576 to 0.2460, and SVM from 0.2563 to 0.2532.

3) No Significant Change::

- Decision Tree maintained similar accuracy levels, indicating limited benefit from this selection method.

C. After Chi-Square + ANOVA Feature Selection

This combined approach leverages statistical techniques to select features that are most relevant with respect to the target variable. The impact here was slightly more promising for some models, though not universally beneficial.

1) Improved Models::

- Logistic Regression once again stood out with the highest overall accuracy (0.2600), outperforming its scores in both the baseline and correlation-based methods. This result suggests that Logistic Regression is highly responsive to statistically-driven feature selection techniques.
- Decision Tree also saw an improvement (from 0.2504 to 0.2559), suggesting that its performance benefits from a more targeted feature subset.
- KNN showed a slight improvement over its previous scores, maintaining decent performance across all settings.

2) Declined Models::

- Gradient Boosting, SVM, and Random Forest experienced accuracy reductions. This trend implies that these models, which are often sensitive to complex feature interactions, may have been negatively affected when potentially subtle but important features were removed.
- For instance, SVM's accuracy dropped further to 0.2457, and Gradient Boosting dipped to 0.2462, suggesting a need for richer feature representations.

3) No Significant Change::

- Naive Bayes, KMeans, and KNN maintained performance in the same range, showing that the new feature set neither significantly improved nor harmed their predictive capabilities.

The table below presents the accuracy values for each classification model across three scenarios: before feature selection, after correlation-based feature selection, and after Chi-Square + ANOVA feature selection.

TABLE III
MODEL PERFORMANCE COMPARISON

Model	Before FS	After Correlation	After Chi2+ANOVA
Naive Bayes	0.2542	0.2555	0.2547
Decision Tree	0.2504	0.2492	0.2559
Random Forest	0.2576	0.2460	0.2519
Gradient Boosting	0.2481	0.2482	0.2462
SVM	0.2563	0.2532	0.2457
Logistic Regression	0.2576	0.2579	0.2600
KMeans	0.2510	0.2536	0.2506
KNN	0.2552	0.2548	0.2553

IV. DISCUSSION

This study evaluated the performance of various classification algorithms on a multiclass dataset consisting of a mix of categorical and numerical features. The models were tested across three stages: before feature selection, after correlation-based feature selection, and after feature selection using Chi-Square and ANOVA methods. The dataset had a relatively balanced distribution among the four classes, as indicated by the similar support values for each class, which ranged between approximately 2390 and 2460 instances. Despite this balance, the classification performance across all models remained notably low, with the highest accuracy achieved being around 0.26.

One possible reason for the consistently poor performance is the inherent complexity and possible lack of strong discriminative patterns within the dataset. Since the dataset comprises both categorical and numerical features, their ability to capture nuanced relationships or patterns that distinguish between classes might be limited if the transformations and encodings do not preserve essential contextual relationships. Even after encoding the variables into numerical representations, the models may have struggled to learn meaningful boundaries.

Additionally, many of the features may not have had a strong direct correlation with the target variable. This is supported by the relatively small gains observed from feature selection. Although feature selection through correlation helped in reducing redundancy, it did not lead to noticeable performance improvements. On the other hand, Chi-Square combined with ANOVA selection improved performance slightly in some models, particularly in logistic regression, but the overall accuracy still remained low.

Another factor contributing to the modest results could be the absence of feature engineering or domain-driven transformation of values. Purely statistical feature selection methods may not be sufficient to reveal underlying structures, especially when feature interactions or latent patterns are critical for accurate predictions. Furthermore, since the dataset contains a combination of categorical and numerical features, many models that rely on numerical gradients or distance measures, such as support vector machines or k-nearest neighbors, may not be operating under optimal conditions if the diverse nature of the data is not fully leveraged.

Interestingly, logistic regression was the best performing model across the three phases. It achieved the highest accuracy of 0.2600 after the application of Chi-Square and ANOVA based feature selection. This result indicates that linear relationships between features and the target were more effectively captured after removing noisy or statistically insignificant attributes. Tree-based models such as decision trees and random forests, which generally perform well on structured datasets, did not show strong performance in this case. This may be attributed to overfitting or an inability to identify high-quality splits due to low feature informativeness.

The KMeans algorithm, an unsupervised method, showed similar accuracy to the other models but failed to provide

meaningful classification, as indicated by the near-zero precision and recall for some classes. This reinforces the fact that clustering models are not designed for label-specific predictions and struggle when class boundaries are not distinct.

Overall, the study reveals that while the dataset is balanced, the classification task remains challenging due to a combination of weak feature relevance, the inherent limitations in how the mixed data types have been transformed, and potentially high overlap in class distributions.

V. Conclusion

The goal of this study was to predict the disease stage in patients suffering from Huntington's disease by leveraging a comprehensive dataset that contains clinical, genetic, and demographic information. Logistic regression emerged as the best performing model among the classifiers tested, particularly when combined with Chi Square and ANOVA based feature selection. Although logistic regression achieved only a modest accuracy of approximately 0.26, it consistently outperformed more complex models. This result indicates that filtering out noisy and statistically insignificant features helped enhance the prediction of the disease stage. The overall low performance of all models suggests that extracting meaningful patterns from the dataset remains challenging.

Future research should focus on developing advanced feature engineering techniques that better capture the underlying biological relationships present in the data. Approaches such as target encoding, frequency encoding, or the use of categorical embeddings from neural networks may be more effective in preserving important contextual information. In addition, a deeper integration of domain knowledge to create higher level features and the exploration of deep learning models designed for tabular data could lead to improved predictive performance in disease stage prediction. Hyperparameter tuning, the use of ensemble techniques, and validation across multiple independent datasets may help improve robustness and reveal additional insights. Ultimately, enhancing performance will require not only more complex models but also the extraction of more informative representations from this diverse and richly annotated dataset.

Acknowledgment

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